Simplified Denitrification Example

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1 Chemistry

We consider a simplified denitrification process in a clean aquifer, where polluted water enters through the inflow. We only consider two redox reactions: aerobic oxidation of DOC and denitrification from nitrate to nitrogen, thus neglecting the entire nitrate reduction sequence. The aqueous species are H^+, Cl^-, NO_3^-, $N_{2(aq)}$, $CH_2O_{(aq)}$, HCO_3^-, $O_{2(aq)}$, CO_3^{2-}, and H_2O. The gases are $CO_{2(g)}$ and $O_{2(g)}$. We assume that there is bicarbonate dissociation in the aquifer. We also assume that water entering the aquifer is in equilibrium with atmospheric O_2 and CO_2.

We consider the following equilibrium reactions.

- Bicarbonate dissociation:

$$HCO_3^- = CO_3^{2-} + H^+$$

- Atmospheric CO_2:

$$H^+ + HCO_3^- = CO_{2(g)} + H_2O$$

- Atmospheric O_2:

$$O_{2(aq)} = O_{2(g)}$$

and kinetic reactions:

- Aerobic oxidation of DOC:

$$CH_2O_{(aq)} + O_{2(aq)} \rightarrow H^+ + HCO_3^-$$

- Denitrification ($NO_3^- \rightarrow N_{2(aq)}$):

$$CH_2O_{(aq)} + 0.8NO_3^- \rightarrow HCO_3^- + 0.2H^+ + 0.4N_{2(aq)} + 0.4H_2O$$

The kinetic reaction rates are governed by the Monod formula (Monod (2012)):

$$\begin{cases} r_{K,1} = \mu_1 \dfrac{[CH_2O_{(aq)}]}{k_{DOC} + [CH_2O_{(aq)}]} \dfrac{[O_{2(aq)}]}{k_{O_2} + [O_{2(aq)}]} \\ \\ r_{K,2} = \mu_2 \dfrac{I_{O_2}}{I_{O_2} + [O_{2(aq)}]} \dfrac{[CH_2O_{(aq)}]}{k_{DOC} + [CH_2O_{(aq)}]} \dfrac{[NO_3^-]}{k_{NO_3^-} + [NO_3^-]} \end{cases}$$

where the brackets indicate concentrations, μ are the maximum reaction rate constants, k are the half-saturation constants, and I are the inhibition constants. The parameter values used in this example are:

Parameter	**Reaction 1**	**Reaction 2**
μ (mol/L/d)	$5 \cdot 10^{-1}$	2
k_{DOC} (mol/L)	10^{-3}	$1.4 \cdot 10^{-3}$
k_{O_2} (mol/L)	10^{-5}	—
$k_{NO_3^-}$ (mol/L)	—	$5.9 \cdot 10^{-3}$
I_{O_2} (mol/L)	—	$2.52 \cdot 10^{-5}$

Table 1: Monod kinetic parameters

The boundary water is in equilibrium with atmospheric O_2 and CO_2, therefore the stoichiometric matrix at the inflow is

$$\mathbf{S}_e^{inf} = \begin{pmatrix} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} & H_2O & CO_{2(g)} & O_{2(g)} \\ -1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

since there are no kinetic reactions. In the domain, there is only one aqueous reactions, therefore the equilibrium stoichiometric matrix in the domain is

$$\mathbf{S}_e^{dom} = \begin{pmatrix} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} & H_2O \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \end{pmatrix}$$

and the kinetic stoichiometric matrix in the domain is

$$\mathbf{S}_K = \begin{pmatrix} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} & H_2O \\ 1 & 0 & 0 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0.2 & 0 & -0.8 & 0.4 & -1 & 1 & 0 & 0 & 0.4 \end{pmatrix}.$$

1.1 Component matrix

Following De Simoni et al. (2005), the component matrix $\mathbf{U}$ is defined such that $\mathbf{U}\mathbf{S}_{e,v}^T = \mathbf{0}$, ensuring that the components $\mathbf{u} = \mathbf{U}\mathbf{c}$ are conservative quantities not affected by equilibrium reactions. Here $\mathbf{S}_{e,v}$ denotes the equilibrium stoichiometric matrix restricted to variable activity species (i.e. excluding constant activity species). In the domain only H_2O has constant activity (no gaseous phase is present), whereas at the inflow H_2O, $CO_{2(g)}$ and $O_{2(g)}$ all have constant activity (the gases are fixed by atmospheric partial pressures). In the domain, $\mathbf{S}_{e,v}$ is

$$\mathbf{S}_{e,v} = \begin{pmatrix} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{pmatrix},$$

obtained from $\mathbf{S}_e^{dom}$ by removing the column corresponding to H_2O.

Splitting $\mathbf{S}_{e,v} = \left(\mathbf{S}'_{e,v} \mid \mathbf{S}''_{e,v}\right)$ into blocks corresponding to primary and secondary variable activity species, and redefining so that $\mathbf{S}''_{e,v} = -\mathbf{I}$, the component matrix takes the form

$$\mathbf{U} = \left(\mathbf{I}_{N_u} \mid {\mathbf{S}'_{e,v}}^T\right), \tag{1.1}$$

where $N_u = N_v - N_r$ is the number of components, N_v is the number of variable activity species and N_r the number of equilibrium reactions.

In the domain, we have $N_v = 8$ variable activity species (H^+, Cl^-, NO_3^-, $N_{2(aq)}$, $CH_2O_{(aq)}$, HCO_3^-, $O_{2(aq)}$, CO_3^{2-}) and $N_r = 1$ equilibrium reaction (bicarbonate dissociation), giving $N_u = 8 - 1 = 7$ primary species. The secondary variable activity species is CO_3^{2-}, whose concentration is determined by equilibrium. The component matrix is therefore $\mathbf{U}^{dom} \in \mathbb{R}^{7\times 8}$:

$$\mathbf{U}^{dom} = \left(\mathbf{I} \mid \mathbf{U}_2^{dom}\right)$$

where

$$\mathbf{U}_2^{dom} = \begin{pmatrix} CO_3^{2-} \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Written out explicitly,

$$
\mathbf{U}^{dom} = \begin{array}{c}
\begin{array}{cccccccc} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \end{array} \\
\begin{pmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{pmatrix}
\end{array},
$$

which yields the conservative components $u_1 = c_{H^+} - c_{CO_3^{2-}}$, $u_2 = c_{Cl^-}$, $u_3 = c_{NO_3^-}$, $u_4 = c_{N_{2(aq)}}$, $u_5 = c_{CH_2O_{(aq)}}$, $u_6 = c_{HCO_3^-} + c_{CO_3^{2-}}$ (total dissolved carbonate), and $u_7 = c_{O_{2(aq)}}$.

The column $\mathbf{U}_2^{dom}$ is obtained from ${\mathbf{S}'_{e,v}}^T$. The bicarbonate dissociation reaction has $\mathbf{S}_{e,v} = (1,0,0,0,0,-1,0,1)$ with respect to the variable activity species $(H^+, Cl^-, NO_3^-, N_{2(aq)}, CH_2O_{(aq)}, HCO_3^-, O_{2(aq)}, CO_3^{2-})$. After redefining $\mathbf{S}''_{e,v} = -\mathbf{I}$ (i.e. dividing by -1 for CO_3^{2-}), the redefined stoichiometric coefficients for the primary species become $\mathbf{S}'_{e,v} = (-1,0,0,0,0,1,0)$ and therefore $\mathbf{U}_2^{dom} = {\mathbf{S}'_{e,v}}^T = (-1,0,0,0,0,1,0)^T$, matching the expression above.

The equilibrium constant for the bicarbonate dissociation at 25°C is $\log_{10} K = -10.3288$, obtained from the thermodynamic database.

Component matrix of the boundary (inflow) water. At the inflow, all three equilibrium reactions apply, so $N_r = 3$. Since H_2O, $CO_{2(g)}$, and $O_{2(g)}$ all have constant activity, $\mathbf{S}_{e,v}^{inf}$ is obtained from $\mathbf{S}_e^{inf}$ by removing these three columns, leaving the same eight variable activity species as in the domain (H^+, Cl^-, NO_3^-, $N_{2(aq)}$, $CH_2O_{(aq)}$, HCO_3^-, $O_{2(aq)}$, CO_3^{2-}), so $N_v = 8$ and $N_u = 8 - 3 = 5$ components. The matrix is

$$
\mathbf{S}_{e,v}^{inf} = \begin{array}{c}
\begin{array}{cccccccc} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \end{array} \\
\begin{pmatrix}
-1 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\
1 & 0 & 0 & 0 & 0 & -1 & 0 & 1
\end{pmatrix}
\end{array},
$$

where the rows correspond to the atmospheric CO_2, atmospheric O_2, and bicarbonate dissociation reactions, in that order. We choose secondary species HCO_3^-, $O_{2(aq)}$, and CO_3^{2-} (one per equilibrium reaction), so the primary species are H^+, Cl^-, NO_3^-, $N_{2(aq)}$, and $CH_2O_{(aq)}$. The secondary block (columns HCO_3^-, $O_{2(aq)}$, CO_3^{2-} of $\mathbf{S}_{e,v}^{inf}$) is

$$
\mathbf{S}_{e,v}^{\prime\prime inf} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{pmatrix},
$$

which is not $-\mathbf{I}$ because the bicarbonate dissociation row has non-zero entries for both HCO_3^- and CO_3^{2-}. Redefining $\mathbf{S}''_{e,v} = -\mathbf{I}$ by premultiplying by $P = -(\mathbf{S}_{e,v}^{\prime\prime inf})^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ – i.e. leaving the atmospheric CO_2 and O_2 rows unchanged and replacing the bicarbonate dissociation row by the atmospheric CO_2 row minus itself – yields the transformed primary block (columns H^+, Cl^-, NO_3^-, $N_{2(aq)}$, $CH_2O_{(aq)}$)

$$
\mathbf{S}_{e,v}^{\prime inf} = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 \end{pmatrix}.
$$

The component matrix at the inflow is $\mathbf{U}^{inf} \in \mathbb{R}^{5\times 8}$:

$$
\mathbf{U}^{inf} = \left(\mathbf{I} \mid \mathbf{U}_2^{inf} \right)
$$

where, with rows corresponding to the primary species H^+, Cl^-, NO_3^-, $N_{2(aq)}$, $CH_2O_{(aq)}$,

$$
\mathbf{U}_2^{inf} = (\mathbf{S}_{e,v}^{\prime inf})^T = \begin{array}{c}
\begin{array}{ccc} HCO_3^- & O_{2(aq)} & CO_3^{2-} \end{array} \\
\begin{pmatrix}
-1 & 0 & -2 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0
\end{pmatrix}
\end{array}.
$$

Written out explicitly in the natural species order,

$$\mathbf{U}^{inf} = \begin{array}{c} \begin{array}{cccccccc} H^+ & Cl^- & NO_3^- & N_{2(aq)} & CH_2O_{(aq)} & HCO_3^- & O_{2(aq)} & CO_3^{2-} \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \end{array}.$$

It is straightforward to verify that $\mathbf{U}^{inf}(\mathbf{S}_{e,v}^{inf})^T = \mathbf{0}$. The five components are $u_1^{inf} = c_{H^+} - c_{HCO_3^-} - 2\,c_{CO_3^{2-}}$ (minus the carbonate alkalinity), $u_2^{inf} = c_{Cl^-}$, $u_3^{inf} = c_{NO_3^-}$, $u_4^{inf} = c_{N_{2(aq)}}$, and $u_5^{inf} = c_{CH_2O_{(aq)}}$. Note that $O_{2(aq)}$ does not appear in any component: its concentration is directly fixed by the atmospheric O_2 equilibrium ($c_{O_{2(aq)}} = K_{O_2}^{-1} a_{O_{2(g)}}$), so it carries no degree of freedom. Likewise, the atmospheric CO_2 equilibrium combined with the bicarbonate dissociation reduces the three carbonate-related species H^+, HCO_3^-, CO_3^{2-} to a single component (the alkalinity). The boundary water therefore has only five independent components, in contrast with the seven components of the domain, reflecting the two additional constraints imposed by equilibrium with the atmospheric gases.

The initial and boundary concentrations we consider are in Table 2:

	initial	inflow
H^+	10^{-7}	10^{-8}
Cl^-	10^{-16}	10^{-3}
NO_3^-	10^{-5}	10^{-3}
$N_{2(aq)}$	10^{-16}	10^{-16}
$CH_2O_{(aq)}$	10^{-5}	$2 \cdot 10^{-3}$
HCO_3^-	10^{-3}	$5.3761 \cdot 10^{-4}$
$O_{2(aq)}$	10^{-6}	$2.5295 \cdot 10^{-4}$

Table 2: Primary species concentrations of initial and boundary waters (in molalities)

2 Transport

We solve the following set of equations:

$$\begin{cases} \phi \dfrac{\partial \mathbf{c}^T}{\partial t} = \mathcal{L}(\mathbf{c}) + \phi \mathbf{r}^T \\ \mathbf{S}_e \log \mathbf{c} = \log \mathbf{K}_e \\ \mathbf{r}_K = \mathbf{r}_K(\mathbf{c}) \end{cases} \tag{2.1}$$

where $\phi \in (0,1]$ is porosity; $\mathbf{c} \in \mathbb{R}^{n_s}$ is the concentration array of all species (n_s is the number of species); $\mathcal{L}(\mathbf{c}) = -\mathbf{q}^T\nabla\mathbf{c}^T + \nabla \cdot \left(\mathbf{D}\nabla\mathbf{c}^T\right) + f_w\,(\mathbf{c}_r - \mathbf{c})$ is the advective form of the transport operator, if we solve transport using the advective ADE, where $\mathbf{q} \in \mathbb{R}^d$ is the Darcy flux (d is the Euclidean dimension, in this case $d = 1$), $\mathbf{D} \in \mathbb{R}^{d\times d}$ is the hydrodynamic dispersion tensor, f_w is the (volumetric) water sink-source, with concentration $\mathbf{c}_r$ if $f_w > 0$ (recharge), or, $\mathbf{c}_r = \mathbf{c}$ if $f_w < 0$ (discharge), or $\mathbf{c}_r = 0$ when $f_w < 0$ represents evaporation; $\mathbf{r} = \mathbf{S}_e^T\mathbf{r}_e + \mathbf{S}_K^T\mathbf{r}_K$ is the contribution of chemical reactions to the mass balance of species (per unit time); $\mathbf{S}_e \in \mathbb{R}^{n_e\times n_s}$ and $\mathbf{S}_K \in \mathbb{R}^{n_K\times n_s}$ are the stoichiometric matrices of equilibrium and kinetic reactions, respectively, whose respective reaction rates are given by $\mathbf{r}_e \in \mathbb{R}^{n_e}$ and $\mathbf{r}_K \in \mathbb{R}^{n_K}$ (n_e and n_K are the number of equilibrium and kinetic reactions, respectively) and $\mathbf{K}_e \in \mathbb{R}^{n_e}$ are the equilibrium constants.

Equation (2.1) needs to be solved subject to appropriate initial ($\mathbf{c}(\mathbf{x},t) = \mathbf{c}(\mathbf{x},0)$ throughout the model domain) and boundary conditions, which we choose to be prescribed mass flux, that is,

$$\begin{cases} \mathbf{n}^T \cdot (\mathbf{q}\mathbf{c} - \mathbf{D}\nabla\mathbf{c}) = q_{inf}\mathbf{c}_{inf} & \text{at the inflow} \\ \mathbf{n}^T \cdot (\mathbf{D}\nabla\mathbf{c}) = 0 & \text{at the outflow} \end{cases}$$

expressing that the flux of solute mass at the inflow is $q_{inf}\mathbf{c}_{inf}$, and that there is no dispersive flux at the outflow.

To solve numerically, we choose a uniform mesh made of 10 cells of length $\Delta x = 0.1$ m (domain length $L = 1$ m). The time step is $\Delta t = 10^{-2}$ days. We use the fully implicit Euler scheme for both transport and reactions. We consider uniform transport properties: $\phi = 0.5$, $q = q_{inf} = 0.2 m/d$, $\alpha_L = 1m$. The dispersion coefficient is $D = \alpha_L q = 0.2 m^2/d$.

Note that while the chemistry described in Section 1 (species, reactions, stoichiometry, equilibrium constants, and Monod parameters) is intrinsic to this denitrification problem, the transport parameters and numerical settings — mesh size, time step, porosity, Darcy flux, dispersivity, boundary condition type, and integration scheme — can be freely modified by the user to suit different physical scenarios, subject to the following stability and accuracy constraints:

- **Grid Péclet number:** $Pe = \frac{q\Delta x}{D} = \frac{\Delta x}{\alpha_L} \leq 2$ to avoid spurious oscillations in centred schemes. In this example, $Pe = 0.1$.
- **Courant number:** $Co = \frac{q\Delta t}{\phi\Delta x} \leq 1$ for time integration. In this example, $Co = 0.2 \cdot 10^{-2}/(0.5 \cdot 0.1) = 0.04$.
- **Von Neumann stability:** For the explicit Euler scheme, the time step must satisfy $\Delta t \leq \frac{\phi(\Delta x)^2}{2D}$ for dispersion stability. The fully implicit Euler scheme is unconditionally stable, so this constraint does not apply.
- **Combined constraint (explicit Euler):** When using explicit time integration, both the Courant and dispersion constraints must be satisfied simultaneously: $\Delta t \leq \min\left(\frac{\phi\Delta x}{q}, \frac{\phi(\Delta x)^2}{2D}\right)$.

The fully implicit Euler scheme used in this example is unconditionally stable, allowing larger time steps at the cost of solving a nonlinear system at each step. However, accuracy still requires Δt small enough to resolve the reaction dynamics.

References

De Simoni, M., Carrera, J., Sanchez-Vila, X., and Guadagnini, A. (2005). A procedure for the solution of multicomponent reactive transport problems. *Water resources research*, 41(11).

Monod, J. (2012). The growth of bacterial cultures. *Annual Review of Microbiology*, 3:371–394.